

Technical Report on the SO-101 Robotic Drawing Project

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Abstract—This report presents an SO-101 robotic oil-pastel drawing system, focusing on workspace calibration, geometric compensation, and tool-wear modeling. A 49-point calibration maps paper coordinates to robot joint configurations, while residual drawing errors inform geometric compensation. A fixed-angle wear model relates oil-pastel mass loss to contact geometry, with empirical stroke-width calibration treated separately. The report summarizes the system setup, methods, and preliminary findings. Future work will evaluate compensation across the workspace, expand wear measurements, and integrate geometric and material-state estimates to improve drawing consistency.

Index Terms—Robotic Drawing, Oil Pastel, Workspace Calibration, Trajectory Error, Error Compensation, Tool Wear, Contact Geometry.

I. INTRODUCTION AND OBJECTIVES

Robotic oil pastel drawing requires converting digital target trajectories into physical strokes on paper. This process involves both the positioning and motion of the robotic arm and the changing condition of the drawing tool. As the oil pastel is consumed during use, the geometry of its tip gradually changes. This may affect the stroke width and the contact condition between the tool and the paper. Therefore, two main technical problems are considered in this project: establishing the mapping between paper coordinates and robot motion, and describing the geometric changes caused by tool wear.

This project uses an SO-101 robotic arm to build an oil pastel drawing platform. Experiments are conducted through trajectory planning, robot execution, and visual observation. From a geometric perspective, target positions on the paper must be converted into corresponding joint configurations of the robot. Workspace calibration provides the basis for this transformation. However, even after calibration, the actual strokes may still show position offsets, orientation errors, and local curvature. These errors may arise from several stages, including coordinate mapping, joint motion, and tool contact. They therefore need to be characterized by comparing the target trajectories with the actual strokes. The conditions under which geometric compensation is effective can then be further evaluated.

From a material perspective, wear of the oil pastel causes the tool geometry to change during use. Even when the same drawing command is repeated, changes in the tool condition may affect the resulting stroke. In this project, the relationship between mass loss and contact geometry is first studied under a fixed tool inclination angle. This reduces the influence of orientation changes on the analysis. The contact area and the

reduction in tool height along the paper normal direction are derived from the geometric model. These quantities are treated separately from the experimentally measured stroke width. The relationship between the model predictions and the observed stroke width must be established through experiments.

Based on these problems, this project has three objectives:

1. Develop the experimental drawing platform. Integrate the robotic arm, oil pastel holder, visual observation system, and trajectory execution process to provide consistent experimental conditions for calibration, error analysis, and wear studies.
2. Characterize geometric errors and investigate compensation methods. Use a 49-point workspace calibration to establish the mapping from paper coordinates to robot joint configurations, analyze the remaining trajectory errors, and evaluate methods that correct drawing targets based on observed errors.
3. Develop a fixed-angle wear model. Describe the relationship between mass loss and tool contact geometry under clearly defined geometric assumptions and operating conditions. Independent measurements of stroke width are then used for empirical calibration and subsequent validation.

II. SYSTEM OVERVIEW AND EXPERIMENTAL SETUP

The system uses an SO-101 robotic arm to build an oil pastel drawing platform. The platform consists of the robotic arm, an end-effector oil pastel holder, a paper workspace, an overhead imaging device, and a computer. The robotic arm is mounted at the edge of the workbench and uses the end-effector holder to secure the oil pastel and execute drawing motions. The paper is fixed on the tabletop, while the imaging device is mounted above the workspace using a support frame to record and observe the drawing results. The computer is responsible for target image processing, stroke planning, execution command generation, and visual analysis. During the experiments, the positions of the paper and the imaging device are kept fixed to provide consistent observation conditions for image comparison across different stages of the drawing process.

The overall system workflow consists of six modules: target image input, stroke planning, geometric compensation, robot execution, visual observation, and error evaluation. The target image defines the desired drawing, while the stroke planning module converts it into an ordered sequence of drawing actions. After the robot executes these actions, the system captures images of the paper to obtain the actual strokes. When



Fig. 1: Hardware setup.

the required conditions are satisfied, the observed errors are used to modify subsequent drawing commands. In the workflow diagram, a complete image task is used to demonstrate the overall planning and drawing process, while straight-line stroke examples are used to illustrate geometric compensation and error analysis.

During the stroke planning stage, the planner generates a sequence of strokes based on the target image and material parameters. Each stroke is represented by a quadratic Bézier curve that defines its geometric shape. It is also associated with an oil pastel color slot and a set of normalized drawing parameters. The planning interface displays the target image, simulated canvas, residual heatmap, and loss curve. These outputs are used to evaluate how well the current stroke sequence approximates the target image. The simulated canvas represents the predicted drawing result under the material model, while the actual result on paper is evaluated using images captured after robot execution.

The geometric compensation module applies bounded corrections to the paper coordinates of future strokes based on previously observed trajectory errors. The system stores both the original reference trajectory and the command trajectory actually sent to the robot. The reference trajectory represents the desired stroke, while the command trajectory includes corrections introduced to compensate for systematic deviations. Maintaining both trajectories allows the error analysis to distinguish between the deviation of the actual stroke from the drawing target and the deviation of the robot execution from the commanded trajectory. The specific compensation method, sources of error, and applicable conditions are discussed in

detail in the next chapter.

During robot execution, trajectories defined in paper coordinates are converted into robot joint commands through workspace calibration. The current calibrated drawing area is 120×120 mm. A 49-point joint-space grid is used to cover this region, and joint-angle interpolation is applied to obtain the joint configurations corresponding to positions along each trajectory. The complete execution procedure also includes lowering the oil pastel onto the paper, drawing along the planned trajectory, lifting the tool, and moving between consecutive strokes. Although these operations are grouped under the robot execution module in the workflow diagram, coordinate mapping remains a necessary step that connects strokes defined on the paper with the motion of the robotic arm.

Visual observation is performed using a checkpoint-based procedure. After completing a group of strokes, the robot lifts the oil pastel and moves to a predefined parking position. An image of the paper is then captured to reduce occlusion caused by the robotic arm. The captured image is corrected for the paper region and registered so that observations from different stages can be represented in a common coordinate frame. The stroke analysis module then extracts information such as the centerline of the actual stroke and compares it with the reference and command trajectories to evaluate positional and shape errors.

Feedback updates are applied only to strokes that have not yet been executed. In observation-only mode, the system records the captured images and analysis results without modifying the drawing commands. When geometric compensation is enabled and the relevant data pass the validity checks, the results of the error analysis are used to update subsequent commands. The drawing task therefore progresses through a sequence of batch execution, parking and observation, error analysis, and conditional updating. This workflow also provides a basis for comparing results before and after compensation, identifying sources of error, and maintaining experimental records.

Fixed-angle oil pastel wear modeling is conducted as an independent analysis module to study the relationship between material loss and tool contact geometry. This chapter mainly describes the information flow and execution structure of the drawing system. The wear model is presented in later chapters. Integration of the wear model with drawing parameter adjustment is treated as part of the next stage of the project.

III. WORKSPACE CALIBRATION AND GEOMETRIC COMPENSATION

A. 49-Point Workspace Calibration

This project uses a 49-point workspace calibration method. The calibration points are uniformly distributed on a 7×7 grid with a spacing of 20 mm, covering a 120×120 mm drawing area. Each grid point stores a corresponding set of robot joint angles. For a target position located inside a grid cell, the execution program performs bilinear interpolation using the joint angles at the four corner points of the cell to generate

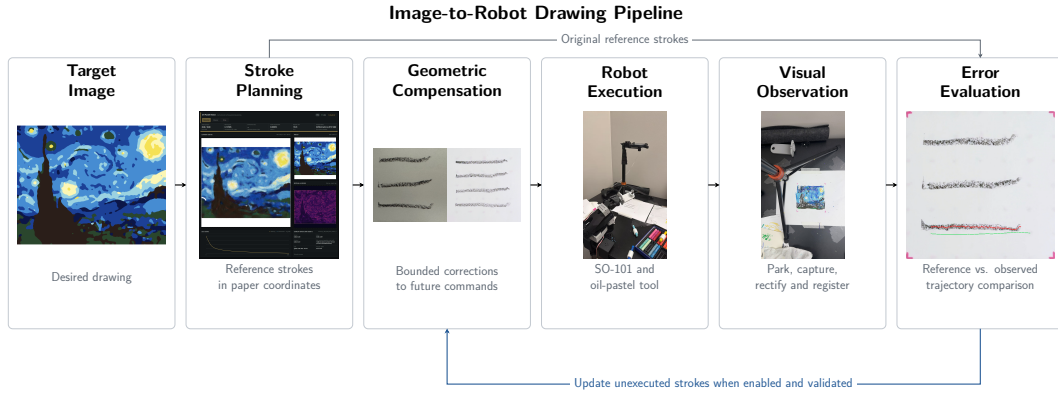


Fig. 2: Overview of the robotic oil-pastel drawing pipeline.

the corresponding joint command. This method establishes a basic mapping between paper coordinates and robot joint configurations. Residual stroke deviations after calibration are further analyzed in later experiments and used for geometric compensation.

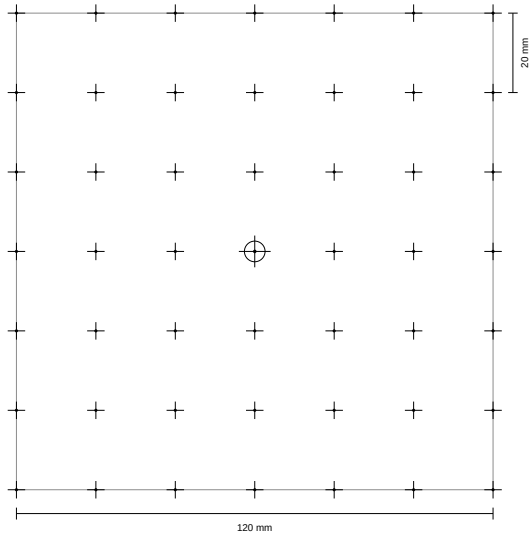


Fig. 3: 49-point workspace calibration.

B. PID Tuning and Joint Error Comparison

In this stage, the PID parameters of each joint were manually adjusted to investigate their effects on joint positioning errors. First, a common setting of $P/I/D = 16/0/32$ was applied to all five actuated joints as the baseline configuration. The PID parameters of the corresponding joints were then adjusted based on the measured joint errors. The wrist-pitch gains were subsequently set to $P/I/D = 16/1/16$ to address vibration during tool lifting and lowering, while the other joints retained their previously tuned gains.

After combining the 120 positioning-error measurements from the five joints, the overall MAE decreased from 0.527° to 0.279° , while the P90 decreased from 1.618° to 0.527° .

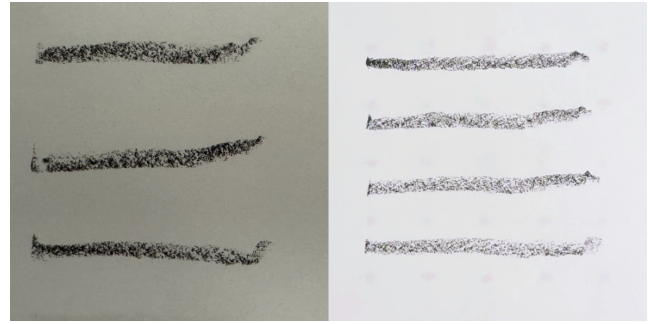


Fig. 4: PID tuning results.

TABLE I: Joint positioning errors before and after PID adjustment. Each value represents the mean absolute angular error calculated from 24 positioning measurements.

Joint	Before Adjustment ($^\circ$)	After Adjustment ($^\circ$)
Base Rotation	0.462	0.429
Shoulder Pitch	0.491	0.123
Elbow Pitch	1.073	0.028
Wrist Pitch	0.377	0.495
Wrist Rotation	0.231	0.319

Most of the improvement came from the shoulder and elbow joints, whereas the MAE of the two wrist joints increased.

C. Visual Feedback Compensation During Drawing

The complete drawing process uses a checkpoint-based visual feedback compensation scheme. After completing at most three strokes, the robotic arm moves to a parking position, and a fixed overhead camera captures the current canvas image. The system first performs paper-region correction and image registration. It then analyzes the difference between consecutive images to identify newly added strokes and estimate their positional deviation and pigment deposition. These observations are used to update the compensation state for subsequent drawing commands that have not yet been executed.

Geometric compensation is mainly used to correct the positional deviation of the actual stroke from the commanded trajectory. The system stores both the original reference trajectory and the trajectory actually sent to the robot. A positional offset is estimated from reliable visual observations, and the remaining strokes are corrected in the direction opposite to the observed offset. Let the reference position be $\mathbf{p}_{\text{ref}}$ and the compensation term be $\Delta\mathbf{p}$. The commanded position is then given by

$$\mathbf{p}_{\text{cmd}} = \mathbf{p}_{\text{ref}} + \Delta\mathbf{p}. \quad (1)$$

The compensation term is constrained by both a maximum correction magnitude and the workspace boundaries to prevent excessive correction caused by observation errors.

For insufficient pigment deposition, the system maintains a separate contact compensation state for each physical oil pastel slot. For the same slot, the system evaluates the average deposition ratio after every three new reliable direct-deposition observations. If the average ratio is below a predefined threshold, the contact offset of the shoulder pitch joint is increased by 1° toward the paper. After the offset reaches $+2^\circ$, if new observations still indicate insufficient deposition, the number of passes over the same stroke is increased progressively, up to a maximum of three passes. Before each reversal, the robot slightly lifts the tool and then lowers it again at the same position.

In addition, a conditional same-color redraw strategy is applied to the final single-pass stroke completed at each checkpoint. A stroke may be redrawn once along a corrected trajectory when its residual relative to the original reference trajectory is approximately a uniform translation, the pigment deposition is insufficient, the observation is considered reliable, and image-based prediction indicates that redrawing is expected to improve the result. After the additional pass, a new image is captured for verification.

IV. FIXED-ANGLE OIL PASTEL WEAR MODELING

A. Model Assumptions and Initial State

The oil pastel is approximated as a uniform-density cylinder. Wear is assumed to occur primarily at the contact end, the worn surface is assumed to remain planar, and the tool orientation is assumed to remain fixed during drawing. The angle between the pastel axis and the paper normal is defined as $\theta = 43^\circ$, corresponding to an angle of 47° between the pastel axis and the paper surface. At this stage, stroke-width calibration is limited to strokes whose drawing direction is parallel to the projection of the pastel axis onto the paper plane.

The oil pastel has a nominal diameter of $D = 9.906$ mm and a length of $L = 70.104$ mm, with the radius defined as $R = D/2$. The initial reference state is established using the net mass of the bare pastel, $m_0 = 8.23$ g, and the first measured stroke width of a new pastel, $w_0 = 4.1$ mm. At this stage, the material density is estimated from the nominal volume of the complete cylinder as

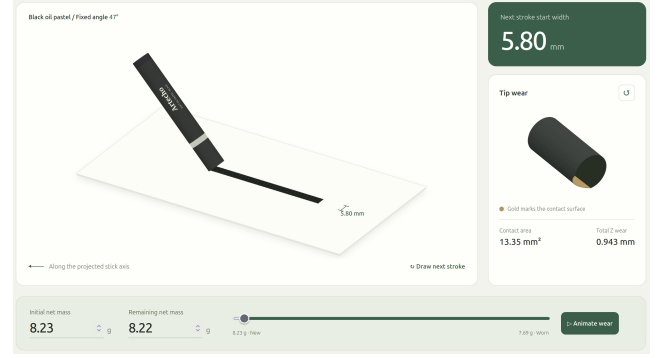


Fig. 5: Oil pastel model.

$$\rho = \frac{m_0}{\pi R^2 L} \approx 1.523 \text{ g/cm}^3. \quad (2)$$

B. Wear Geometry Estimated from Mass Loss

Let h denote the circular-segment height of the equivalent contact region on a circular cross-section perpendicular to the oil pastel axis. The corresponding projected area is

$$S(h) = R^2 \arccos\left(\frac{R-h}{R}\right) - (R-h)\sqrt{2Rh-h^2}. \quad (3)$$

The model is defined over the range $h_0 \leq h \leq 2R$, which represents the wear process from the initial reference state to the point at which a complete oblique section is first formed. Since the contact area changes with the wear state, the lost material volume is calculated by integration. For the current net mass m_t , the cumulative mass loss is related to the wear state by

$$\Delta m_t = m_0 - m_t = \rho \tan \theta \int_{h_0}^{h_t} S(u) du. \quad (4)$$

After numerically solving this relation for the current wear state h_t , the geometric contact area A_t and the paper-normal recession relative to the initial state can be calculated as

$$A_t = \frac{S(h_t)}{\cos \theta}, \quad z_{\text{wear},t} = (h_t - h_0) \sin \theta. \quad (5)$$

Here, $S(h_t)$ represents the projected area on the cross-section perpendicular to the oil pastel axis, while A_t represents the inclined contact area in the model. A positive value of $z_{\text{wear},t}$ indicates recession of the contact end along the paper-normal direction due to material wear.

V. EVALUATION METRICS

This work defines evaluation metrics at three levels: paper-space trajectory accuracy, prediction error of the wear model, and full-drawing image quality.



Fig. 6: Final result.

A. Stroke Position Accuracy

Stroke position accuracy is evaluated using the deviation of the observed stroke centerline from the original reference trajectory. First, paper markers are used for image correction and scale conversion, and the extracted stroke centerline is mapped to paper coordinates in millimeters. The reference trajectory is then sampled at a fixed interval. The normal error at the i th sample point is defined as

$$e_i^\perp = \mathbf{n}_i^\top (\mathbf{p}_i^{\text{obs}} - \mathbf{p}_i^{\text{ref}}), \quad (6)$$

where $\mathbf{p}_i^{\text{obs}}$ and $\mathbf{p}_i^{\text{ref}}$ denote the corresponding observed and reference positions, respectively, and $\mathbf{n}_i$ is the unit normal vector of the reference trajectory at that point.

The mean absolute error and the 90th percentile of the absolute normal error are used as the primary evaluation metrics:

$$\text{MAE}_\perp = \frac{1}{N} \sum_{i=1}^N |e_i^\perp|, \quad P90_\perp = Q_{0.90} \left(\{|e_i^\perp|\}_{i=1}^N \right). \quad (7)$$

Both metrics are reported in millimeters. $\text{MAE}_\perp$ represents the average positional deviation, while $P90_\perp$ reflects larger local deviations. Results before and after compensation are evaluated using the same original reference trajectory, sampling interval, and image-processing procedure. Centerline overlay plots are also used to visualize changes in trajectory error. In addition, the two-dimensional deviations at the stroke start and end points are reported as supplementary metrics to reflect positioning errors along the drawing direction.

B. Wear Model Prediction Accuracy

The primary evaluation metric for the wear model is the difference between the predicted stroke width and the measured stroke width. For the k th validation state, let $\hat{w}_k$ denote the stroke width predicted by the model and w_k denote the independently measured stroke width. The mean absolute error of the stroke width is defined as

$$\text{MAE}_w = \frac{1}{M} \sum_{k=1}^M |\hat{w}_k - w_k|. \quad (8)$$

This metric is reported in millimeters. A smaller value indicates that the predicted stroke width is closer to the measured result and therefore corresponds to higher prediction accuracy.

C. Overall Drawing Similarity

Overall drawing similarity is evaluated by comparing the final captured canvas with the target image. Weighted L1 error is used as the primary evaluation metrics. This combination follows the evaluation approach used in IMPASTO for assessing physical strokes and multi-step painting results.

The weighted L1 error measures color differences at the pixel level while assigning greater importance to the target content region. Let the captured image and the target image be denoted by I_{obs} and I_{tar} , respectively, with RGB values normalized to $[0, 1]$. The weighted L1 error is defined as

$$E_{\text{wL1}} = \frac{\sum_{p \in \Omega} \omega_p \|I_{\text{obs}}(p) - I_{\text{tar}}(p)\|_1}{3 \sum_{p \in \Omega} \omega_p}. \quad (9)$$

Here, Ω denotes the evaluation region, and ω_p is the weight assigned to pixel p . The weights are determined in advance from the target image and kept fixed across all comparisons. A nonzero weight is retained for the background region so that undesired strokes outside the target content region are also included in the evaluation. A smaller value of this metric indicates a smaller pixel-level difference between the captured and target images.

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